LAB 5: Non-linear Equations

Objective: To obtain the numerical solution of any non-linear equation.

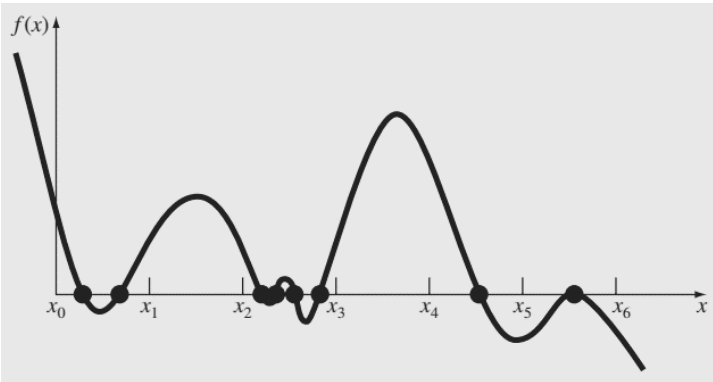
Theory: Root of the equation f(x) = 0 is to make a plot of the function and observe where it crosses the x axis. The two major classes of methods available are distinguished by the type of initial guess. They are

1. Bracketing methods - As the name implies, these are based on two initial guesses that "bracket" the root that is, are on either side of the root.
2. Open methods - These methods can involve one or more initial guesses, but there is no need for them to bracket the root.

f (x) changed sign on opposite sides of the root. In general, if f(x) is real and continuous in the interval from x_l to x_u and $f(x_l)$ and $f(x_u)$ have opposite signs, that is,

$$f(x_l)\,f(x_u) < 0$$

then there is at least one real root between x_l and x_u.



The Bisection method –

Example - To determine the mass of the bungee jumper with a drag coefficient of 0.25 kg/m to have a velocity of 36 m/s after 4 s of free fall. Note: The acceleration of gravity is 9.81 m/s2

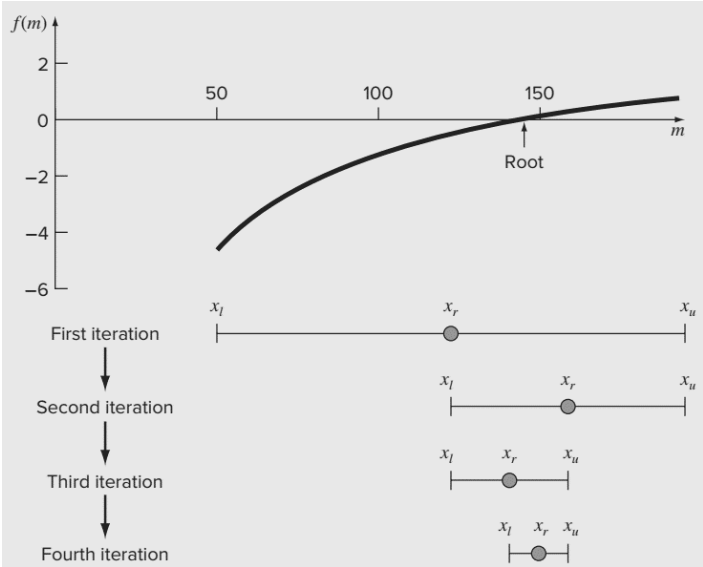
$$f(m) = \sqrt{\frac{gm}{c_d}}\tanh\left(\sqrt{\frac{gc_d}{m}}\,t\right) - v(t)$$

We can see that the function changes sign between values of 50 and 200. The plot obviously suggests better initial guesses, say 140 and 150, but for illustrative purposes let's assume we don't have the benefit of the plot and have made conservative guesses. Therefore, the initial estimate of the root x_r lies at the midpoint of the interval,

$$x_r = \frac{50 + 200}{2} = 125$$

$$|\varepsilon_t| = \left|\frac{142.7376 - 125}{142.7376}\right| \times 100\% = 12.43\%$$

$$f(50)\,f(125) = -4.579(-0.409) = 1.871$$



At this point, the new interval extends from $x_l = 125$ to $x_u = 200$. A revised root estimate can then be calculated as

$$x_r = \frac{125 + 200}{2} = 162.5$$

which represents a true percent error of $|\varepsilon_t| = 13.85\%$. The process can be repeated to obtain refined estimates. For example,

$$f(125)\,f(162.5) = -0.409(0.359) = -0.147$$

Therefore, the root is now in the lower interval between 125 and 162.5. The upper bound is redefined as 162.5, and the root estimate for the third iteration is calculated as

$$x_r = \frac{125 + 162.5}{2} = 143.75$$

which represents a percent relative error of $\varepsilon_t = 0.709\%$. The method can be repeated until the result is accurate enough to satisfy your needs.

Stopping criteria,

$$|\varepsilon_a| = \left|\frac{x_r^{\text{new}} - x_r^{\text{old}}}{x_r^{\text{new}}}\right| 100\%$$

Iteration	x_l	x_u	x_r	$\lvert\varepsilon_a\rvert$ (%)	$\lvert\varepsilon_t\rvert$ (%)
1	50	200	125		12.43
2	125	200	162.5	23.08	13.85
3	125	162.5	143.75	13.04	0.71
4	125	143.75	134.375	6.98	5.86
5	134.375	143.75	139.0625	3.37	2.58
6	139.0625	143.75	141.4063	1.66	0.93
7	141.4063	143.75	142.5781	0.82	0.11
8	142.5781	143.75	143.1641	0.41	0.30

MATLAB code –

```
clc;close all; clear all;
m = linspace(0,250,50);
g = 9.8; cd = 0.25; t = 4; v = 36;
ml =50;
mu =200;
err= 0.01;
 iter = 1;
 while abs(mu-ml)> err
    mr = (ml+mu)/2;

    yl = sqrt(g*ml/cd)*tanh(sqrt(g*cd/ml)*t) - v;
    yr = sqrt(g*mr/cd)*tanh(sqrt(g*cd/mr)*t) - v;
    yu = sqrt(g*mu/cd)*tanh(sqrt(g*cd/mu)*t) - v;

   if yl*yr <0
      mu=mr;
   else
      ml=mr;
   end
  iter = iter+1
end
fprintf('root of equation = %f',mr);
```

Other methods – False position, Newton-Raphson method etc.

Example- Find the roots of equation – $y(x) = e^{0.2x} - e^{-0.8x} - 2$ using bisection method. Take the initial guesses as root may lie between 0 and 4.

MATLAB routine for finding roots is as follows-

```
clc;close all; clear all;
x = linspace(0,10,50);
y = exp(0.2*x)-exp(-0.8*x)-2;
xl =0;
xu =4;
err= 0.01;
 iter = 1;
 while abs(xu-xl)> err
     xr = (xl+xu)/2;
     yl = exp(0.2*xl)-exp(-0.8*xl)-2;
     yr = exp(0.2*xr)-exp(-0.8*xr)-2;
     yu = exp(0.2*xu)-exp(-0.8*xu)-2;
    if yl*yr <0
        xu=xr;
    else
        xl=xr;
    end
   iter = iter+1;
end
fprintf('root of equation = %f',xr);
plot(x,y);
hold on;
hline = refline([0 0]);
```

Assignment -

One numerical as suggested by Faculty.

Sample numerical.

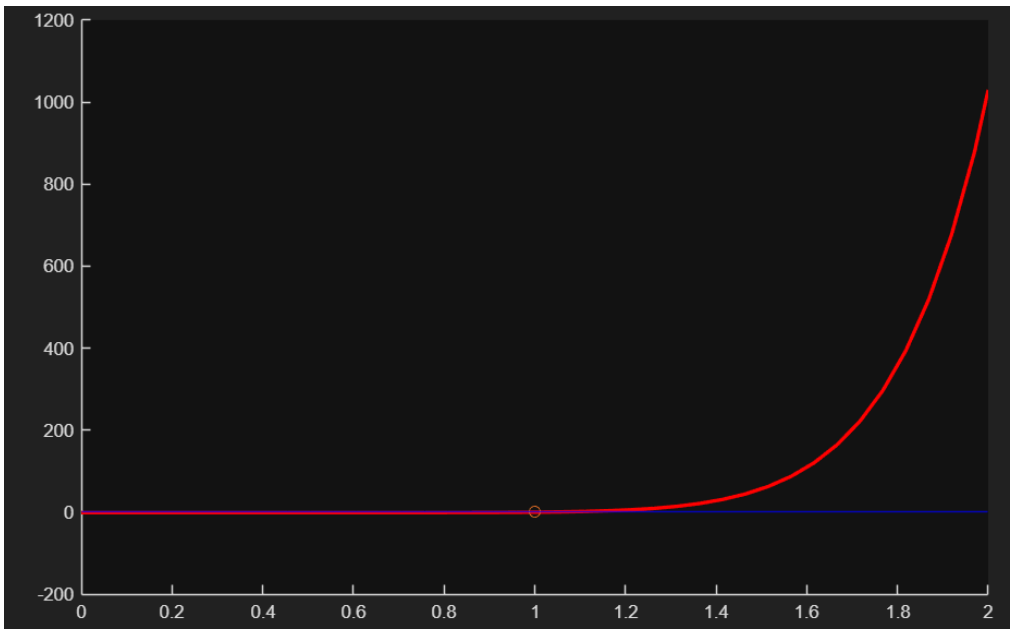
Using Bisection method, locate the root of –

$$F(x) = x^{10} - 1, \text{ between } x = 0 \text{ to } 1.3$$

Experiment 5: Non Linear Equations

Matlab Code

```
clc;close all; clear all;
x = linspace(0,5,100);
y = x.^10 - 1;
xl = 0;
xu = 1.3;

err = 0.01;
iter = 1;


while abs(xu-xl)>err
    xr = (xu+xl)/2;
    yl = xl^10 -1;
    yr = xr^10 -1;
    yu = xu^10 -1;

    if yu*yr < 0
        xl = xr;
    else
        xu = xr;
    end
    iter = iter+1;

end

yline(0,'b','LineWidth',0.5);
hold on;
plot(x,y, 'r', 'LineWidth',2);
plot(xr, yr, '-o')

xlim([0 2]);

fprintf("The root of this function is: %f", xr)

```

Output and Plot

The root of this function is: 1.000391